GSE 518
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Problem Set 1

Turn in any 3 out of the 5 problems below. However, you are strongly encouraged to work on all the problems before the midterm, since the material they cover will be part of the midterm. Please let me know if you have any questions.

1. Prove the following identities:

   (a) $(A \cap B) \cap A \backslash B = \phi$, where ϕ is the empty set.

   (b) $A \cup B = A \cup \left(B \cap A^C\right)$

   (c) If $B \subset A$, then $A = A \backslash B \cup B$.

2. Two six-sided dice are thrown sequentially, and the face values are recorded. List the elements that make up the following events and give the number of elements in each of them.

   (a) $A =$ the sum of the two values is at least 5.

   (b) $B =$ the value of the first die is higher than that of the second.

   (c) $C =$ the first value is four.

   (d) $A \cap C$.

   (e) $B \cup C$.

   (f) $A \cap (B \cup C)$.

3. Prove that $P(A \cap B) = 1 - P\left(A^C\right) - P\left(B^C\right) + P\left(A^C \cap B^C\right)$. (*Hint:* Try using the Implications of Kolmogorov's Axioms along with DeMorgan's law.) (*Good to know:* This may be the most difficult problem in this problem set.)

4. Prove that if $A \subset B$, then $P(A) \leq P(B)$. (*Hint:* Try using Kolmogorov's Axioms along with the result in Problem 1(c).)

5. Problem 1.13 from CB.

1.9 Prove the general version of DeMorgan's Laws. Let $\{A_\alpha: \alpha \in \Gamma\}$ be a (possibly uncountable) collection of sets. Prove that

(a) $(\cup_\alpha A_\alpha)^c = \cap_\alpha A_\alpha^c$. (b) $(\cap_\alpha A_\alpha)^c = \cup_\alpha A_\alpha^c$.

1.10 Formulate and prove a version of DeMorgan's Laws that applies to a finite collection of sets $A_1, \ldots, A_n$.

1.11 Let S be a sample space.

(a) Show that the collection $\mathcal{B} = \{\emptyset, S\}$ is a sigma algebra.

(b) Let $\mathcal{B} = \{\text{all subsets of } S, \text{ including } S \text{ itself}\}$. Show that $\mathcal{B}$ is a sigma algebra.

(c) Show that the intersection of two sigma algebras is a sigma algebra.

1.12 It was noted in Section 1.2.1 that statisticians who follow the deFinetti school do not accept the Axiom of Countable Additivity, instead adhering to the Axiom of Finite Additivity.

(a) Show that the Axiom of Countable Additivity implies Finite Additivity.

(b) Although, by itself, the Axiom of Finite Additivity does not imply Countable Additivity, suppose we supplement it with the following. Let $A_1 \supset A_2 \supset \cdots \supset A_n \supset \cdots$ be an infinite sequence of nested sets whose limit is the empty set, which we denote by $A_n \downarrow \emptyset$. Consider the following:

Axiom of Continuity: If $A_n \downarrow \emptyset$, then $P(A_n) \to 0$.

Prove that the Axiom of Continuity and the Axiom of Finite Additivity imply Countable Additivity.

1.13 If $P(A) = \frac{1}{3}$ and $P(B^c) = \frac{1}{4}$, can A and B be disjoint? Explain.

1.14 Suppose that a sample space S has n elements. Prove that the number of subsets that can be formed from the elements of S is 2^n.

1.15 Finish the proof of Theorem 1.2.14. Use the result established for $k = 2$ as the basis of an induction argument.

1.16 How many different sets of initials can be formed if every person has one surname and

(a) exactly two given names? (b) either one or two given names?

(c) either one or two or three given names?

(Answers: (a) 26^3 (b) $26^3 + 26^2$ (c) $26^4 + 26^3 + 26^2$)

1.17 In the game of dominoes, each piece is marked with two numbers. The pieces are symmetrical so that the number pair is not ordered (so, for example, $(2, 6) = (6, 2)$). How many different pieces can be formed using the numbers $1, 2, \ldots, n$?
(Answer: $n(n+1)/2$)

1.18 If n balls are placed at random into n cells, find the probability that exactly one cell remains empty.
(Answer: $\binom{n}{2} n!/n^n$)

1.19 If a multivariate function has continuous partial derivatives, the order in which the derivatives are calculated does not matter. Thus, for example, the function $f(x, y)$ of two variables has equal third partials

$$\frac{\partial^3}{\partial x^2 \partial y} f(x, y) = \frac{\partial^3}{\partial y \partial x^2} f(x, y).$$

(a) How many fourth partial derivatives does a function of three variables have?

(b) Prove that a function of n variables has $\binom{n+r-1}{r}$ rth partial derivatives.

1.20 My telephone rings 12 times each week, the calls being randomly distributed among the 7 days. What is the probability that I get at least one call each day?
(Answer: .2285)